

Logical Equivalences Involving Quantifiers

01. $\forall x \in D, P(x) \wedge Q(x) \equiv (\forall x \in D, P(x)) \wedge (\forall x \in D, Q(x))$.

02. $\forall x \in D, P(x) \vee Q(x) \equiv (\forall x \in D, P(x)) \vee (\forall x \in D, Q(x))$.

Negation of conjunction and disjunction with open sentences

(i) $\neg(\forall x \in D, P(x) \wedge Q(x)) \equiv \exists x \in D, \neg(P(x) \wedge Q(x))$
 $\equiv \exists x \in D, (\neg P(x)) \vee (\neg Q(x))$

(ii) $\neg(\forall x \in D, P(x) \vee Q(x)) \equiv \exists x \in D, \neg(P(x) \vee Q(x))$
 $\equiv \exists x \in D, (\neg P(x)) \wedge (\neg Q(x))$

(iii) $\neg(\exists x \in D, P(x) \wedge Q(x)) \equiv \forall x \in D, \neg(P(x) \wedge Q(x))$
 $\equiv \forall x \in D, (\neg P(x)) \vee (\neg Q(x))$

(iv) $\neg(\exists x \in D, P(x) \vee Q(x)) \equiv \forall x \in D, \neg(P(x) \vee Q(x))$
 $\equiv \forall x \in D, (\neg P(x)) \wedge (\neg Q(x))$

Implication and open sentences

$$\forall x \in D, P(x) \Rightarrow Q(x).$$

Here are some common ways this is done.

- Given $x \in D$, if $P(x)$, then $Q(x)$.
- Let $x \in D$. If $P(x)$, then $Q(x)$.
- If x is an arbitrary element of D satisfying $P(x)$, then $Q(x)$.
- If x is an element of D satisfying $P(x)$, then $Q(x)$.

$$\exists x \in D, P(x) \Rightarrow Q(x).$$

The words we associate with $\exists$ are: "for some", "exists", "at least", etc.

- Consider the conditional quantification: $\forall x \in D, P(x) \Rightarrow Q(x)$

- Then we defined

➤ **Converse:** $\forall x \in D, Q(x) \Rightarrow P(x)$

➤ **Contrapositive:** $\forall x \in D, \neg Q(x) \Rightarrow \neg P(x)$

➤ **Inverse:** $\forall x \in D, \neg P(x) \Rightarrow \neg Q(x)$

- **Note:** A conditional proposition is logically equivalent to its contrapositive.

- **Negation of Conditional Quantification**

$$\neg(\forall x \in D, P(x) \Rightarrow Q(x)) \equiv \exists x_0 \in D, \neg(P(x_0) \Rightarrow Q(x_0))$$

$$\equiv \exists x_0 \in D, P(x_0) \wedge \neg Q(x_0)$$

Multiple quantifiers and negation

- $\neg(\forall x \in S, \forall y \in T, \exists z \in U, P(x, y, z)) \equiv \exists x \in S, \exists y \in T, \forall z \in U, \neg P(x, y, z)$.
- $\neg(\exists x \forall y \exists z P(x, y, z)) \equiv \forall x \exists y_0 \forall z \neg P(x, y_0, z)$

2. Introduction to Proofs

- **Theorem:** A statement that can be shown to be true (a fact or result).
- **Proposition:** A smaller or less important theorem.
- **Axiom or Postulate:** A statement that is assumed to be true without proof.
- **Lemma:** A less important theorem used as a stepping stone in the proof of another theorem.
- **Corollary:** A less important theorem that follows directly from a previously proven larger theorem.
- **Conjecture:** A statement proposed as true but not yet proven.
- ✗ • **Proof:** A valid argument that establishes the truth of a theorem or any statement that can be true or false.

Definition: An argument (logical deduction) is an assertion that a given set of statements $p_1, p_2, \dots, p_n$, called *hypotheses* or *premises*, yield another statement q , called the *conclusion*. Such a logical deduction is denoted by:

$$p_1, p_2, \dots, p_n \vdash q \text{ or}$$

$$\begin{array}{c} p_1 \\ p_2 \\ \vdots \\ p_n \\ \hline \therefore q \end{array}$$

Note:

- An argument is a series of statements starting from **hypothesis** (premises/assumptions) and ending with the **conclusion**.
- From the definition, an argument might be **valid** or **invalid**.

Definition: Valid Arguments (Proofs)

An argument $p_1, p_2, \dots, p_n \vdash q$ is said to be valid if q is true whenever all the premises $p_1, p_2, \dots, p_n$ are true; otherwise it is invalid.

i.e. Argument $p_1, p_2, \dots, p_n \vdash q$ is valid if and only if " $(p_1 \wedge p_2 \wedge \dots \wedge p_n) \Rightarrow q$ " is tautology.

Note

In mathematics, the word *proof* is used to mean simply a valid argument. Many proofs involve more than two premises and a conclusion.

Example1:

If 870,232 is divisible by 6, then it is divisible by 3.

870,232 is not divisible by 3.

Therefore, 870,232 is not divisible by 6.

Let p : 870232 is divisible by 6
 q : 870232 is divisible by 3

$$\begin{array}{l} p \Rightarrow q \\ \sim q \\ \hline \therefore \sim p \end{array}$$

$p \Rightarrow q, \sim q \vdash \sim p$
 $\Rightarrow p \Rightarrow q, \sim q \vdash \sim p$ is valid
 $(p \Rightarrow q, \sim q) \Rightarrow \sim p$ is tautology

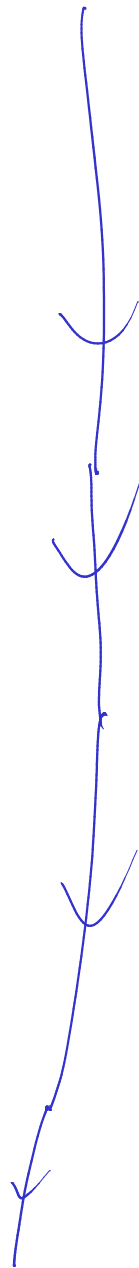
Suppose $p \Rightarrow q$ is T — (i)
 $\sim q$ is T — (ii)
 By (i) q is F — (iii)

By (i) & (iii), p must be F
 $\therefore \sim p$ is T
 $\therefore$ the argument is valid

$$((p \Rightarrow q) \wedge (\neg q)) \Rightarrow (\neg p)$$

p	q	$p \Rightarrow q$	$\neg q$	$(p \Rightarrow q) \wedge (\neg q)$	$\neg p$	$((p \Rightarrow q) \wedge (\neg q)) \Rightarrow (\neg p)$
T	T	T	F	F	F	T
F	T	T	F	F	T	T
T	F	F	T	F	F	T
F	F	T	T	T	T	T

Since $((p \Rightarrow q) \wedge (\neg q)) \Rightarrow (\neg p)$ is tautology, the argument $p \Rightarrow q, \neg q \vdash \neg p$ is valid



Ans: let p : - - - -
 q : - - - -
 r : - - - -

Suppose $p \Rightarrow q$ is T (i)
 and $q \Rightarrow r$ is T (ii)
 When p is F: conclusion $p \Rightarrow r$ is T
 When p is T: q must be T (by (i))
 $\therefore r$ must be T (by (ii))

Then the given argument is of the form $p \Rightarrow q, q \Rightarrow r \vdash p \Rightarrow r$
 In this case we have p is T and r is T
 $\therefore p \Rightarrow r$ is T

Example 2:

If 18,486 is divisible by 18, then 18,486 is divisible by 9.

If 18,486 is divisible by 9, then the sum of the digits of 18,486 is divisible by 9.

Therefore, If 18,486 is divisible by 18, then the sum of the digits of 18,486 is divisible by 9.

$p \Rightarrow q$
 $q \Rightarrow r$
 $\therefore p \Rightarrow r$

$p \Rightarrow r$ is T
 $\therefore$ the argument is valid

Example 3:

If a number is divisible by 6, then it is divisible by 3

120 is divisible by 3.

Therefore, 120 is divisible by 6

$\forall n \in \mathbb{Z}, p(n) \Rightarrow q(n)$
 $p(n)$: n is divisible by 6
 $q(n)$: n is divisible by 3
 $\forall n \in \mathbb{Z}, p(n) \Rightarrow q(n)$
 $q(120)$
 $\therefore p(120)$

Example 4:

If a number is divisible by 9, then the sum of its digits is divisible by 9.

The sum of digits of 145 is 10, which is not divisible by 9.

Therefore, 145 is not divisible by 9.

$\forall n \in \mathbb{Z}, p(n) \Rightarrow q(n)$
 $\sim q(145)$
 $\therefore \sim p(145)$

Let $p(n)$: n is divisible by 9
 $q(n)$: sum of the digits of n is divisible by 9
 $(\forall n \in \mathbb{Z}, p(n) \Rightarrow q(n)), \sim q(145) \vdash \sim p(145)$

Rules of inferences

Inference rules or logical deductions are used to prove new propositions using previously proved ones.

A fundamental inference rule is modus ponens. This rule says that " p implies q together with a p " is a proof of q .

Modes Ponens

$p \Rightarrow q$ (Premise 1)

p (Premise 2)

$\therefore q$ (Conclusion)

i.e. Modus Ponens is a valid argument " $(p \Rightarrow q), p \vdash q$ " equivalently, " $((p \Rightarrow q) \wedge p) \Rightarrow q$ " is tautology.

Other two impotent inference rules are given below:

Modes Tollens

$p \Rightarrow q$ (Premise 1)

$\neg q$ (Premise 2)

$\therefore \neg p$ (Conclusion)

$\sim q \Rightarrow \sim p$
 $\sim q$
 $\therefore \sim p$

i.e. Modus Tollens is a valid argument " $(p \Rightarrow q), \neg q \vdash \neg p$ " equivalently, " $((p \Rightarrow q) \wedge \neg q) \Rightarrow \neg p$ " is tautology.

Chain Rule (Principle of Syllogism)

$p \Rightarrow q$ (Premise 1)

$q \Rightarrow r$ (Premise 2)

$\therefore p \Rightarrow r$ (Conclusion)

i.e. Principle of Syllogism is a valid argument " $(p \Rightarrow q), (q \Rightarrow r) \vdash (p \Rightarrow r)$ " equivalently,

" $(p \Rightarrow q) \wedge (q \Rightarrow r) \Rightarrow (p \Rightarrow r)$ " is tautology.

Example 1: Proof of $\sqrt{3} < \sqrt{5}$.

$p \Rightarrow r$
 $r \Rightarrow s$
 $s \Rightarrow t$
 $t \Rightarrow u$

$p \Rightarrow s$
 $s \Rightarrow u$
 $\therefore p \Rightarrow u$

" $3 < 5 \Rightarrow (3 - 5) < 0$

$$(3 - 5) < 0 \Rightarrow (\sqrt{3} - \sqrt{5})(\sqrt{3} + \sqrt{5}) < 0$$

$$(\sqrt{3} - \sqrt{5})(\sqrt{3} + \sqrt{5}) < 0 \Rightarrow (\sqrt{3} - \sqrt{5}) < 0$$

$$(\sqrt{3} - \sqrt{5}) < 0 \Rightarrow \sqrt{3} < \sqrt{5}$$

$\therefore "3 < 5 \Rightarrow \sqrt{3} < \sqrt{5}"$ (By Principle of Syllogism)

Now we have $3 < 5 \Rightarrow \sqrt{3} < \sqrt{5}$ and

$3 < 5$ is T

$$\frac{p \Rightarrow q}{p} \therefore q$$

$\therefore \sqrt{3} < \sqrt{5}$ (By Modes Ponens)

$$\frac{p \Rightarrow q}{\sim q} \therefore \sim p$$

Example 2: Proof of $\sqrt{2} + \sqrt{3} < 11$.

$$\sqrt{2} + \sqrt{3} \geq 11 \Rightarrow 6 \geq (58)^2$$

$$\neg(6 \geq (58)^2)$$

$\therefore \sqrt{2} + \sqrt{3} < 11$. (By Modes Tollens)

Rules of Inference for Quantified Statements

- **Universal Instantiation**

$$\forall x, P(x)$$

$$\therefore P(c)$$

Example: $\forall n \in \mathbb{Z}, n^2 \geq 0$.

$$\therefore (-13751)^2 \geq 0.$$

- **Universal Generalization**

$P(c)$ for an arbitrary c

$$\therefore \forall x, P(x)$$

Example: $a^2 > 0$ for an arbitrary non-zero real number a .

$$\therefore \forall x \in \mathbb{R} \setminus \{0\}, x^2 > 0.$$

- **Existential Instantiation**

$$\exists x, P(x)$$

$$\therefore P(c) \text{ for some } c$$

Example: $|0| = -0, |-1| = -(-1), |-\sqrt{5}| = -(-\sqrt{5}) \dots$ etc.

$$\exists x \in \mathbb{R}, |x| = -x.$$

$$\therefore |c| = -c \text{ for some real number } c.$$

- **Existential Generalization**

$P(c)$ for some c

$$\therefore \exists x, P(x)$$

Example: $1^2 = 1$

i.e. $c^2 = c$ for some real number c .

$$\therefore \exists x \in \mathbb{R}, x^2 = x.$$

Suppose $\forall x, p(x) \Rightarrow q(x)$ is T. — (i)
al $\forall x, p(x)$ is T. — (ii)
Let c be an arbitrary element
By (i) $p(c) \Rightarrow q(c)$ is T — (iii)
By (ii) $p(c)$ is T — (iv)
 $\therefore q(c)$ is T (by (iii) & (iv))

$Q(c)$ is true for arbitrary c
 $\therefore \underline{\forall x, Q(x)}$ is $\top$
 $\therefore$ the argument is valid

Example: Determine the validity of the following argument:

$\forall x, P(x) \Rightarrow Q(x)$ (Premise 1)

$\forall x, P(x)$ (Premise 2)

$\therefore \forall x, Q(x)$ (Conclusion)

" $(\forall x, P(x) \Rightarrow Q(x)), (\forall x, P(x)) \vdash \forall x, Q(x)$ "
Is this valid argument?

End of D7